Math 661 Applied Statistics – Module 11 Assignment

Figures

Exercise 7.18 plots:

Figure 1A. Histogram of MPG (Exercise 7.18)

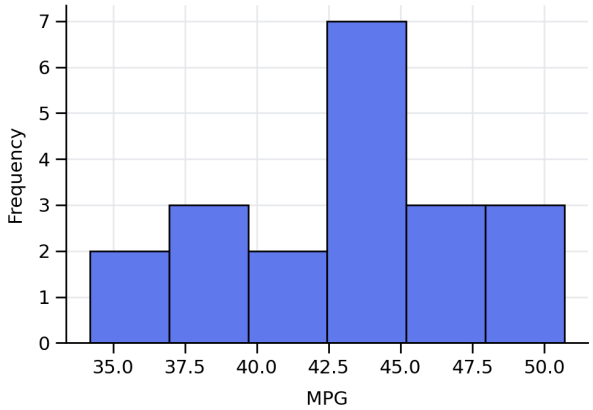


Figure 1B. Boxplot of MPG (Exercise 7.18)

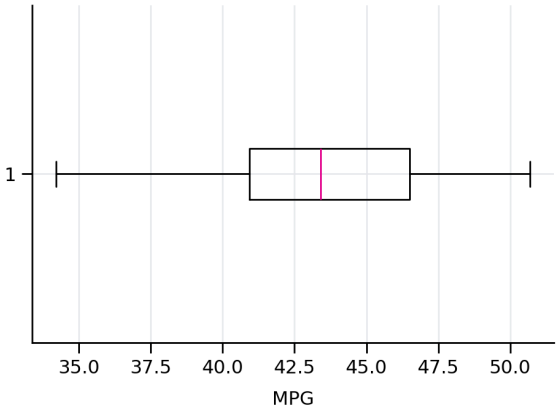


Figure 1C. Normal Q-Q Plot of MPG

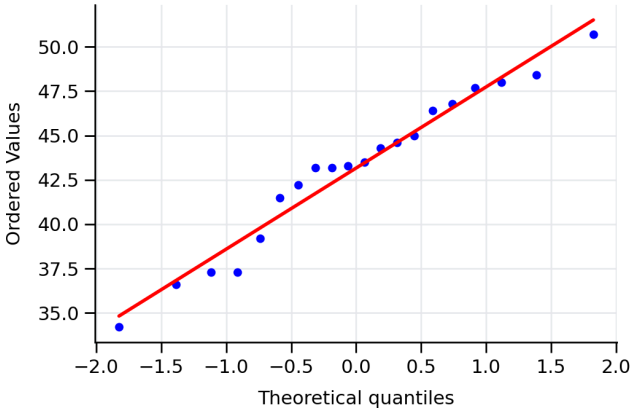
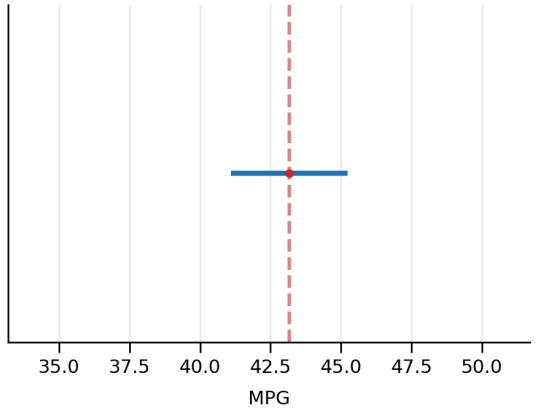


Figure 1D. 95% CI for Mean MPG



Exercise 7.18 — Fuel efficiency *t*test

The 20 mpg values are:

41.5, 50.7, 36.6, 37.3, 34.2, 45.0, 48.0, 43.2, 47.7, 42.2,
43.2, 44.6, 48.4, 46.4, 46.8, 39.2, 37.3, 43.5, 44.3, 43.3

Let x= mpg for one fill-up.

7.18(a) Describe the distribution using graphical methods. Is a Normal-based method appropriate?

Visual evidence

From the histogram, boxplot, and normal Q–Q plot (see the first figure), the distribution:

- is **unimodal**,
- is centered in the low-to-mid 40s,
- has **no strong outliers**,
- shows **only slight skewness**.

Five-number summary

Sorted data:

34.2, 36.6, 37.3, 37.3, 39.2, 41.5, 42.2, 43.2, 43.2, 43.3,

43.5, 44.3, 44.6, 45.0, 46.4, 46.8, 47.7, 48.0, 48.4, 50.7

Five-number summary:

Statistic	Value
Minimum	34.2
Q_1	40.35
Median	43.4
Q_3	46.60
Maximum	50.7

Outlier check

$$IQR = Q_3 - Q_1 = 46.6 - 40.35 = 6.25$$

Outlier fences:

$$Q_1 - 1.5(IQR) = 40.35 - 1.5(6.25) = 30.975$$
$$Q_3 + 1.5(IQR) = 46.6 + 1.5(6.25) = 55.975$$

All observations are between 30.975 and 55.975, so there are **no outliers**.

Conclusion for 7.18(a)

It is reasonable to use **Normal-based methods** (specifically, a one-sample *t*interval) because the data are roughly symmetric/unimodal with no outliers and no severe non-Normal features.

$$A\ one - sample\ t\ procedure\ is\ appropriate.$$

7.18(b) Find the mean, standard deviation, standard error, and margin of error for 95% confidence

Step 1: Sample mean

$$\bar{x} = \frac{863.4}{20} = 43.17$$
$$\bar{x} = 43.17\ mpg$$

Step 2: Sample standard deviation

$$s = 4.415\ mpg$$

Step 3: Standard error

$$SE_{\bar{x}} = \frac{s}{\sqrt{n}} = \frac{4.415}{\sqrt{20}} = 0.987$$
$$SE = 0.987\ mpg$$

Step 4: Margin of error

Since σis unknown, use a t-distribution with

$$df = n - 1 = 19$$

For 95% confidence:

$$t^* = 2.093$$

So:

$$ME = t^* \cdot SE = 2.093(0.987) = 2.066$$
$$ME \approx 2.07\ mpg$$

7.18(c) Report the 95% confidence interval for μ, the mean mpg

$$\bar{x} \pm ME = 43.17 \pm 2.07$$

Therefore,

$$(43.17 - 2.07,\ 43.17 + 2.07 = (41.10,\ 45.24)$$
$$95\%\ CI\ for\ \mu = (41.10,\ 45.24)\ mpg$$

Interpretation

We are 95% confident that the true mean fuel efficiency for this vehicle lies between **41.10 mpg** and **45.24 mpg**.

Summary table for Exercise 7.18

Quantity	Value

Sample size n	20
Mean $\bar{x}$	43.17
Standard deviation s	4.415
Standard error	0.987
t^*(95%, df = 19)	2.093
Margin of error	2.066
95% CI	(41.10, 45.24)

Exercise 7.25 plots:

Figure 2A. MPG by Fill-up (Exercise 7.25)

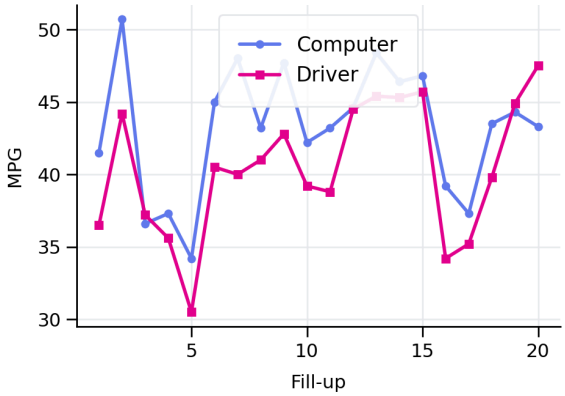


Figure 2B. Paired Plot of Computer vs Driver

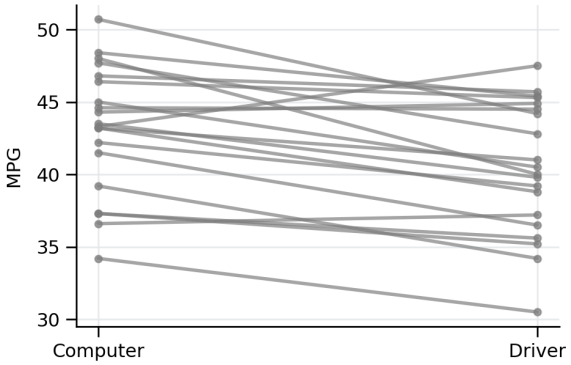


Figure 2C. Histogram of Paired Differences (Computer - Driver)

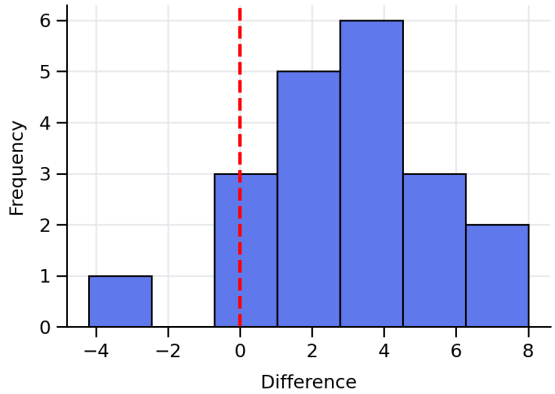
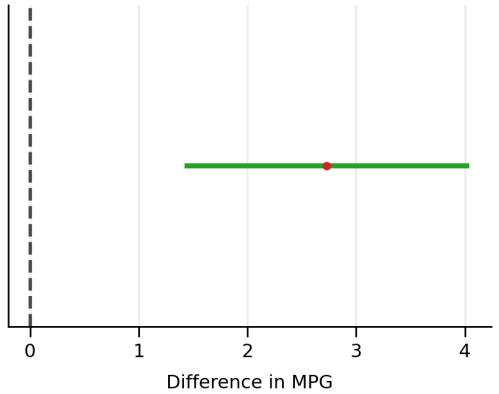


Figure 2D. 95% CI for Mean Difference (Computer - Driver)



Exercise 7.25 — Fuel efficiency comparison test

For the same 20 fill-ups, the **computer** and the **driver** both recorded mpg. Because the two measurements are from the **same fill-up**, this is a **matched pairs** problem.

Let

$$d = Computer - Driver$$

Data table and paired differences

Fill-up	Computer	Driver	Difference d
1	41.5	36.5	5.0
2	50.7	44.2	6.5
3	36.6	37.2	-0.6
4	37.3	35.6	1.7
5	34.2	30.5	3.7
6	45.0	40.5	4.5
7	48.0	40.0	8.0

8	43.2	41.0	2.2
9	47.7	42.8	4.9
10	42.2	39.2	3.0
11	43.2	38.8	4.4
12	44.6	44.5	0.1
13	48.4	45.4	3.0
14	46.4	45.3	1.1
15	46.8	45.7	1.1
16	39.2	34.2	5.0
17	37.3	35.2	2.1
18	43.5	39.8	3.7
19	44.3	44.9	-0.6
20	43.3	47.5	-4.2

Differences:

5.0, 6.5, −0.6, 1.7, 3.7, 4.5, 8.0, 2.2, 4.9, 3.0,
4.4, 0.1, 3.0, 1.1, 1.1, 5.0, 2.1, 3.7, −0.6, −4.2

7.25(a) State the hypotheses

Let μ_d be the true mean of the paired differences $d = Computer - Driver$.

The driver wants to know whether the two calculations are **different**, so this is a two-sided test.

$$H_0: \mu_d = 0$$
$$H_a: \mu_d \neq 0$$

7.25(b) Carry out the test at significance level 0.05. Give the P-value and interpret the result.

Step 1: Summary statistics for the differences

- Number of pairs:

$$n = 20$$

- Mean difference:

$$\bar{d} = 2.73$$

- Standard deviation of differences:

$$s_d = 2.802$$

- Standard error:

$$SE_{\bar{d}} = \frac{s_d}{\sqrt{n}} = \frac{2.802}{\sqrt{20}} = 0.626$$

Step 2: Test statistic

$$t = \frac{\bar{d} - 0}{SE_{\bar{d}}} = \frac{2.73}{0.626} = 4.358$$

Degrees of freedom:

$$df = n - 1 = 19$$

So,

$$t = 4.358, df = 19$$

Step 3: P-value

For a two-sided test:

$$P \approx 0.00034$$

Step 4: Decision

Since

$$0.00034 < 0.05$$

we **reject** H_0.

Conclusion

There is **very strong evidence** that the computer and driver calculations do not give the same average mpg. Because the mean difference is positive ($\bar{d} = 2.73$), the computer's mpg values are, on average, about **2.73 mpg higher** than the driver's.

$$\textit{The computer tends to report higher mpg than the driver.}$$

Additional confidence interval (useful support)

A 95% confidence interval for μ_d is:

$$\bar{d} \pm t^* \left(\frac{s_d}{\sqrt{n}} \right)$$

with $df = 19$ and $t^* = 2.093$.

$$2.73 \pm 2.093(0.626) = 2.73 \pm 1.31 \quad (4.04)$$

This interval does not include 0, which supports the rejection of H_0.

Summary table for Exercise 7.25

Quantity	Value
Number of pairs n	20
Mean difference $\bar{d}$	2.73
SD of differences s_d	2.802
SE of $\bar{d}$	0.626
Test statistic t	4.358
Degrees of freedom	19
P-value	0.00034
95% CI for μ_d	(1.42, 4.04)

Exercises 7.86–7.87 plots:

Figure 3A. Group Values by Pair (Exercise 7.86)

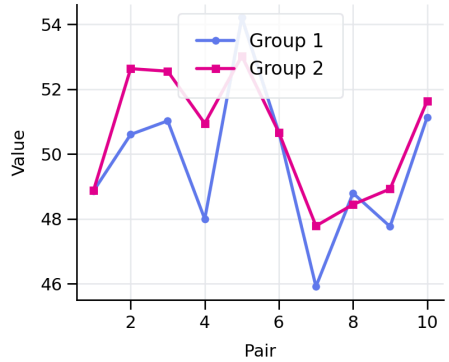


Figure 3B. Paired Plot for Group 1 vs Group 2

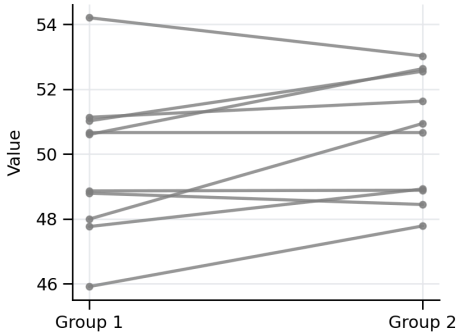


Figure 3C. Histogram of Paired Differences (Group 1 - Group 2)

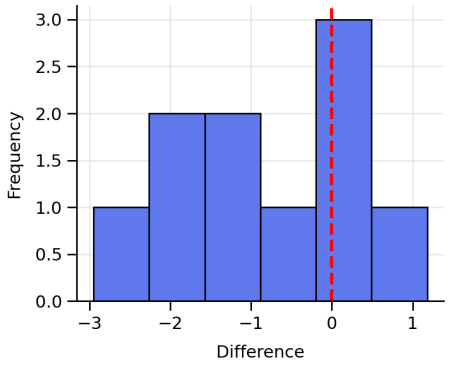
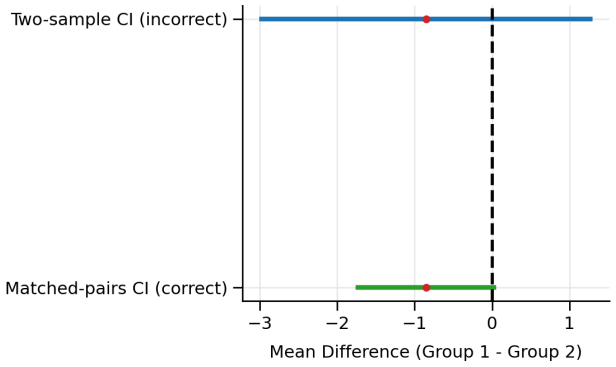


Figure 3D. 95% CI Comparison (Exercise 7.87)



Exercise 7.86 — Two-sample *t*test versus matched pairs *t*test

The data were collected in **pairs**, but part (a) asks us to pretend they are independent and compare that with the correct matched-pairs analysis.

Data

Pair	Group 1	Group 2
1	48.86	48.88
2	50.60	52.63
3	51.02	52.55
4	47.99	50.94
5	54.20	53.02
6	50.66	50.66
7	45.91	47.78
8	48.79	48.44
9	47.76	48.92
10	51.13	51.63

Let

$$d = Group\,1 - Group\,2$$

so the differences are:

−0.02, −2.03, −1.53, −2.95, 1.18, 0.00, −1.87, 0.35, −1.16, −0.50

7.86(a) Incorrect analysis: treat as a two-sample problem

Step 1: Compute the sample means

For Group 1:

$$\bar{x}_1 = \frac{496.92}{10} = 49.692$$

For Group 2:

$$\bar{x}_2 = \frac{505.45}{10} = 50.545$$

Estimated difference:

$$\bar{x}_1 - \bar{x}_2 = 49.692 - 50.545 =- 0.853$$

Step 2: Compute the sample variances

$$s_1^2 = 5.37264, s_2^2 = 3.70316$$

Step 3: Standard error

Using the two-sample $t$formula:

$$SE = \sqrt{\frac{s_1^2}{n_1} + \frac{s_2^2}{n_2}}$$

with $n_1 = n_2 = 10$,

$$SE = \sqrt{\frac{5.37264}{10} + \frac{3.70316}{10}}$$

$$SE = \sqrt{0.537264 + 0.370316} = \sqrt{0.907580} = 0.95267$$

Step 4: Test statistic

We test

$$H_0: \mu_1 - \mu_2 = 0 vs H_a: \mu_1 - \mu_2 \neq 0$$

The test statistic is

$$t = \frac{(\bar{x}_1 - \bar{x}_2) - 0}{SE}$$

$$t = \frac{-0.853}{0.95267} = -0.895$$

Step 5: Degrees of freedom and P-value

Using the conservative IPS rule:

$$df = min(n_1 - 1, n_2 - 1) = min(9, 9) = 9$$

So the two-sided P-value is

$$P \approx 0.394$$

Conclusion for 7.86(a)

$$t = -0.895, df = 9, P \approx 0.394$$

Since $P > 0.05$, we fail to reject H_0. If we (incorrectly) treat the samples as independent, we do **not** find a significant difference.

7.86(b) Correct analysis: matched pairs $t$test

Since the observations are naturally paired, the proper analysis uses the differences d.

Step 1: Mean of the differences

$$\bar{d} = \frac{-8.53}{10} = -0.853$$
$$\bar{d} = -0.853$$

Step 2: Variance and standard deviation of the differences

$$s_d^2 = 1.61067$$
$$s_d = \sqrt{1.61067} = 1.26912$$

Step 3: Standard error

$$SE_{\bar{d}} = \frac{s_d}{\sqrt{n}} = \frac{1.26912}{\sqrt{10}} = 0.40128$$

Step 4: Test statistic

We test

$$H_0: \mu_d = 0 vs H_a: \mu_d \neq 0$$

The matched-pairs $t$statistic is:

$$t = \frac{\bar{d} - 0}{SE_{\bar{d}}} = \frac{-0.853}{0.40128} = -2.125$$

Step 5: Degrees of freedom and P-value

$$df = n - 1 = 9$$

Two-sided P-value:

$$P \approx 0.0625$$

Conclusion for 7.86(b)

$$t = -2.125, df = 9, P \approx 0.0625$$

At the 5% level, we still fail to reject H_0, but the evidence against H_0 is **much stronger** than in the incorrect two-sample analysis.

7.86(c) Describe the differences in the two test results

Comparison table

Method	Estimate	Standard Error	t	df	P-value
Two-sample (incorrect)	-0.853	0.9527	-0.895	9	0.394
Matched pairs (correct)	-0.853	0.4013	-2.125	9	0.0625

Discussion

1. The estimated difference is the **same** in both approaches:

$$\bar{x}_1 - \bar{x}_2 = \bar{d} = -0.853$$

2. The standard error is much smaller for the matched-pairs analysis:

$$0.4013 \; vs \; 0.9527$$

3. Because the matched-pairs analysis has a smaller standard error, it produces:
   - a larger $|t|$,
   - a smaller P-value,
   - stronger evidence against H_0.

Why this happens

The pairing removes much of the pair-to-pair variability. The two values in each pair tend to move together, so their **difference** is less variable than the raw observations. Ignoring the pairing wastes information.

Conclusion for 7.86(c)

The matched-pairs test is the correct method and is more powerful. The incorrect two-sample test makes the difference look less important because it ignores the matched design.

Exercise 7.87 — Two-sample $t$test versus matched pairs t test, continued

This exercise compares **95% confidence intervals** for the mean difference using the two approaches.

7.87(a) 95% CI using the two-sample $t$interval

From 7.86(a):

$$\bar{x}_1 - \bar{x}_2 = -0.853$$
$$SE = 0.95267$$

Using the IPS conservative degrees of freedom:

$$df = 9$$

For 95% confidence and $df = 9$,

$$t^* = 2.262$$

So the margin of error is

$$ME = t^* \cdot SE = 2.262(0.95267) = 2.155$$

Therefore the interval is

$$-0.853 \pm 2.155$$
$$(-3.008,\ 1.302)$$

7.87(b) 95% CI using the matched pairs $t$interval

From 7.86(b):

$$\bar{d} = -0.853, SE_{\bar{d}} = 0.40128, df = 9$$

Again, for 95% confidence with $df = 9$,

$$t^* = 2.262$$

So the margin of error is

$$ME = 2.262(0.40128) = 0.908$$

Thus the interval is

$$-0.853 \pm 0.908$$
$$(-1.761,\ 0.055)$$

7.87(c) Compare the estimates (centers) and margins of error

Centers

Both intervals have the same center:

$$-0.853$$

This is because both methods estimate the same mean difference, and

$$\bar{x}_1 - \bar{x}_2 = \bar{d}$$

Margins of error

- Two-sample interval:

$$ME = 2.155$$

- Matched-pairs interval:

$$ME = 0.908$$

So the matched-pairs interval is much narrower.

Visual comparison

```
Two-sample CI:    (-3.008 ------------------- -0.853 ------------------- 1.302)

Matched-pairs CI: (-1.761 -------- -0.853 -------- 0.055)
```

Interpretation

Both intervals include 0, so both are compatible with "no true difference" at the 95% level. However, the matched-pairs interval is much more precise, showing that the paired design substantially reduces variability.

Conclusion for 7.87(c)

The major difference is **precision**. The matched-pairs approach has the same center but a much smaller margin of error because it correctly uses the within-pair information.

Exercise 7.3 — Finding the critical value t^*

The critical value t^* for a confidence interval for a population mean is taken from the t-**distribution** with

$$df = n - 1$$

and the confidence interval is

$$\bar{x} \pm t^*\left(\frac{s}{\sqrt{n}}\right).$$

7.3(a) A 95% confidence interval based on $n = 18$ observations

Step 1: Degrees of freedom

$$df = n - 1 = 18 - 1 = 17$$

Step 2: Confidence level

For a 95% confidence interval:

$$\alpha = 1 - 0.95 = 0.05$$

Because the interval is two-sided, each tail has area

$$\alpha/2 = 0.025$$

So we use

$$t^* = t_{0.975,17}$$

Step 3: Critical value

From Table D:

$$t^* \approx 2.110$$

7.3(b) A 95% confidence interval from an SRS of 26 observations

Step 1: Degrees of freedom

$$df = 26 - 1 = 25$$

Step 2: Critical value

For 95% confidence:

$$t^* = t_{0.975,25}$$

From Table D:

$$t^* \approx 2.060$$

7.3(c) A 90% confidence interval from a sample of size 26

Step 1: Degrees of freedom

$$df = 26 - 1 = 25$$

Step 2: Confidence level

For a 90% confidence interval:

$$\alpha = 1 - 0.90 = 0.10, \alpha/2 = 0.05$$

So the critical value is

$$t^* = t_{0.95,25}$$

From Table D:

$$t^* \approx 1.708$$

7.3(d) How does margin of error depend on confidence level and sample size?

The margin of error is

$$ME = t^*\left(\frac{s}{\sqrt{n}}\right)$$

From this formula:

Effect of confidence level

If the confidence level increases, then t^* increases, so the margin of error increases.

$$Higher\ confidence\ level\ \Rightarrow\ larger\ margin\ of\ error$$

For example, with $n = 26$:

- 95% confidence: $t^* = 2.060$
- 90% confidence: $t^* = 1.708$

Since $2.060 > 1.708$, the 95% interval is wider.

Effect of sample size

As n increases, $\sqrt{n}$ increases, so

$$\frac{s}{\sqrt{n}}$$

gets smaller. Therefore the margin of error decreases.

$$Larger\ sample\ size\ \Rightarrow\ smaller\ margin\ of\ error$$

Conclusion

The margin of error:

- **increases** as the confidence level increases,
- **decreases** as the sample size increases.

Exercise 7.4 — Distribution of the t statistic

The problem states that the sample size is $n = 20$.

Step 1: Degrees of freedom

$$df = n - 1 = 20 - 1 = 19$$

Step 2: Significance level

The test is **two-sided** at the **1% level**, so

$$\alpha = 0.01$$

For a two-sided test, each tail has area

$$\alpha/2 = 0.005$$

So the critical value is

$$t^* = t_{0.995,19}$$

From Table D:

$$t^* \approx 2.861$$

Step 3: Rejection region

For a two-sided $t$test, reject H_0if the test statistic falls in either extreme tail:

$$t <- 2.861 or t > 2.861$$

Equivalently,

$$|t| > 2.861$$

Sketch of the distribution

```
                    t distribution (df = 19)

                         ^
```

```
                        |
                      .-'-.
                    .'     '.
                  .'         '.
--------------.'-----------'.--------------------> t
              -2.861     0     2.861

        Reject H0                  Reject H0
       (area = 0.005)            (area = 0.005)
```

The center area is 0.99, and the two tails together have area 0.01.

Interpretation

Values of t near 0 are consistent with the null hypothesis. Very large positive or negative values give strong evidence against H_0. Therefore, at the 1% level for a two-sided test, we reject only when the test statistic is less than -2.861 or greater than 2.861.

Consolidated Results Table

Exercise	Key Result
7.3(a)	$t^* = 2.110$ for 95% CI, $n = 18$
7.3(b)	$t^* = 2.060$ for 95% CI, $n = 26$
7.3(c)	$t^* = 1.708$ for 90% CI, $n = 26$
7.3(d)	Higher confidence ⇒ larger MOE; larger n ⇒ smaller MOE
7.4	Reject H_0 if $t < -2.861$ or $t > 2.861$
7.18	$\bar{x} = 43.17,\ s = 4.415,\ SE = 0.987,\ ME = 2.07,\ 95\% CI = (41.10, 45.24)$
7.25	Matched-pairs test: $t = 4.358,\ df = 19,\ P = 0.00034$; computer reports higher mpg on average
7.86(a)	Incorrect two-sample test: $t = -0.895,\ df = 9,\ P = 0.394$
7.86(b)	Correct matched-pairs test: $t = -2.125,\ df = 9,\ P = 0.0625$
7.87(a)	Two-sample 95% CI: $(-3.008,\ 1.302)$
7.87(b)	Matched-pairs 95% CI: $(-1.761,\ 0.055)$
7.87(c)	Same center, but matched-pairs interval is much narrower

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Final Overall Conclusion

These exercises illustrate several core ideas from inference for means:

1. **Critical values** depend on confidence level and degrees of freedom.
2. **Higher confidence** produces a **larger margin of error**, while **larger sample size** produces a **smaller margin of error**.
3. For one-sample inference (Exercise 7.18), the t-interval is appropriate when the data show no strong skewness or outliers.
4. For paired data (Exercises 7.25, 7.86, and 7.87), the **matched-pairs $t$procedure** is the correct analysis. It is usually more informative than treating the observations as independent because it reduces variability and leads to more precise inference.